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Report

Laboratory Work No. 4

Operating Systems

Topic: The xv6 Shell from the Outside

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INTRODUCTION

Figure note. Terminal-style figures below are reconstructions from the supplied report, saved files and terminal record. They are not original screenshots of the running virtual machine.

The user interface of xv6 is a single program, `user/sh.c`, of roughly 500 lines. In this laboratory work that shell is used next to bash, so that it becomes visible, by experiment, what a shell has to do and what bash adds on top of it. A small program that prints the arguments it receives is then used to find out who expands `*` and who removes quotes. Before that, the `sleep` program from Laboratory Work No. 2 is checked against the current xv6, where the system call `sleep()` was renamed to `pause()`.

The purpose of the work is to make `sleep` build on the current xv6, to compare the xv6 shell with bash command by command, to show with the program `args` which component turns `*.c` into file names and strips the quotes, and to read the matching code in `user/sh.c`.

The report follows the four parts of the handout; the first number of every figure is the number of the part. The experiments were done on Ubuntu 26.04.1 LTS running in an Oracle VirtualBox virtual machine, under the user account `dalvgame`. The bash side of the comparison was run in an empty folder `~/shtest` that contains one file, `README`. The xv6 side was run at the `$` prompt of xv6-riscv started with `make qemu` in `~/xv6-riscv`; the tools were installed and checked with the course script `os2026-setup.sh` (QEMU 10.2, compiler `riscv64-linux-gnu-gcc`). The deliverables `sleep.c`, `args.c`, `args_linux.c` and the `Makefile` are stored in the `lab4/` folder of the course repository together with this report.

1 FIXING SLEEP FOR THE CURRENT XV6

This chapter corresponds to Part 1 of the handout. In August 2025 the xv6 authors renamed the system call `sleep()` to `pause()`, so the `sleep` program of Laboratory Work No. 2 may no longer build.

1.1 Checking the Name Used by the Tree

```
cd ~/xv6-riscv
grep -n "pause\|sleep" user/user.h
25:int pause(int);
```

The evidence for this exercise is shown in Figure 1.1.



Figure 1.1 – The grep result in user.h

Comment and observation

The grep found only one declaration, on line 25 of `user/user.h`: `int pause(int);`. There is no declaration of `sleep`, so this tree uses the new name and the call in `user/sleep.c` has to be `pause`.

1.2 The Updated sleep.c

```
#include "kernel/types.h"
#include "user/user.h"

int
main(int argc, char *argv[])
{
  if(argc != 2){
    fprintf(2, "usage: sleep <ticks>\n");
    exit(1);
  }
  pause(atoi(argv[1]));   // was sleep() before Aug 2025
  exit(0);
}
```

The evidence for this exercise is shown in Figure 1.2.



Figure 1.2 – The updated sleep.c

Comment and observation

The program requires exactly one argument; otherwise it writes `usage: sleep <ticks>` to descriptor 2 (standard error) and exits with status 1. The argument is converted with `atoi`, the process is paused for that many clock ticks, and it exits with status 0. The Makefile contains the line `$U/_sleep\` in the `UPROGS` list (it starts with a TAB, ends with a backslash and has nothing after it), so `mkfs` copies the program into the xv6 file system.

1.3 Build, Boot and Test

```
make qemu
$ sleep 20
$ sleep
usage: sleep <ticks>
```

The evidence for this exercise is shown in Figure 1.3.

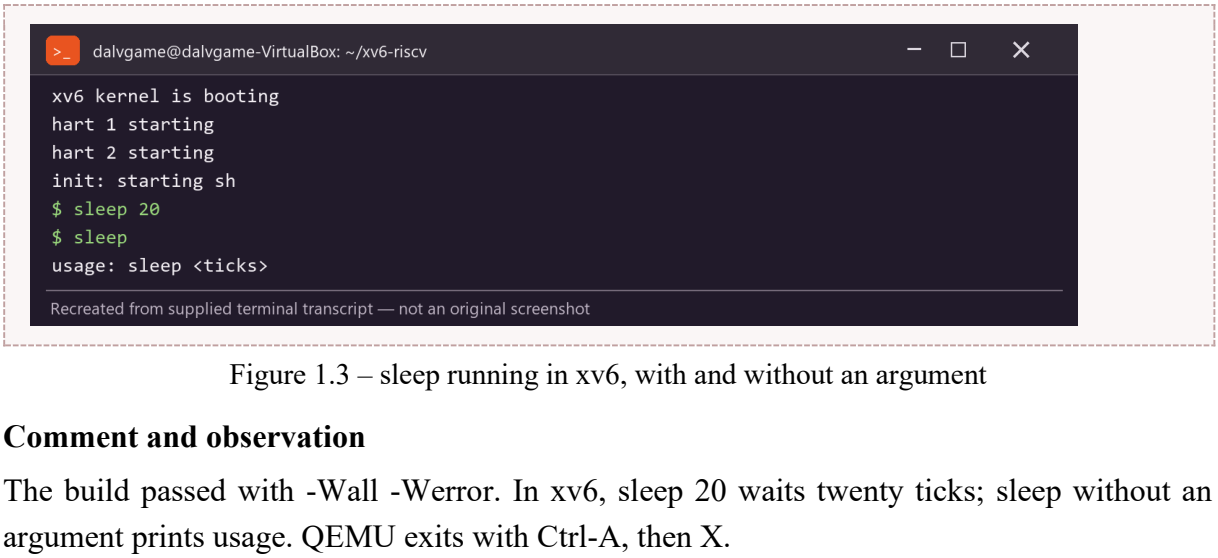


Figure 1.3 – sleep running in xv6, with and without an argument

Comment and observation

The build passed with -Wall -Werror. In xv6, sleep 20 waits twenty ticks; sleep without an argument prints usage. QEMU exits with Ctrl-A, then X.

2 BASH VERSUS THE XV6 SHELL

This chapter corresponds to Part 2 of the handout. Every command was run twice: in bash, in the empty folder `~/shtest` that contains a file `README`, and at the xv6 `$` prompt.

2.1 The Commands Run in bash

```
mkdir ~/shtest && cd ~/shtest && touch README
echo hello > f ; cat < f
ls | grep READ
(echo a; echo b) | wc
echo one ; echo two
echo bg &
ls *
echo $HOME
echo "a b"
mkdir d ; cd d ; ls .. ; cd ..
```

Output:

```
hello
README
      2       2       4
one
two
[1] 7059
bg
README  f
[1]+  Done                    echo bg
/home/dalvgame
a b
README  d  f
```

The evidence for this exercise is shown in Figure 2.1.

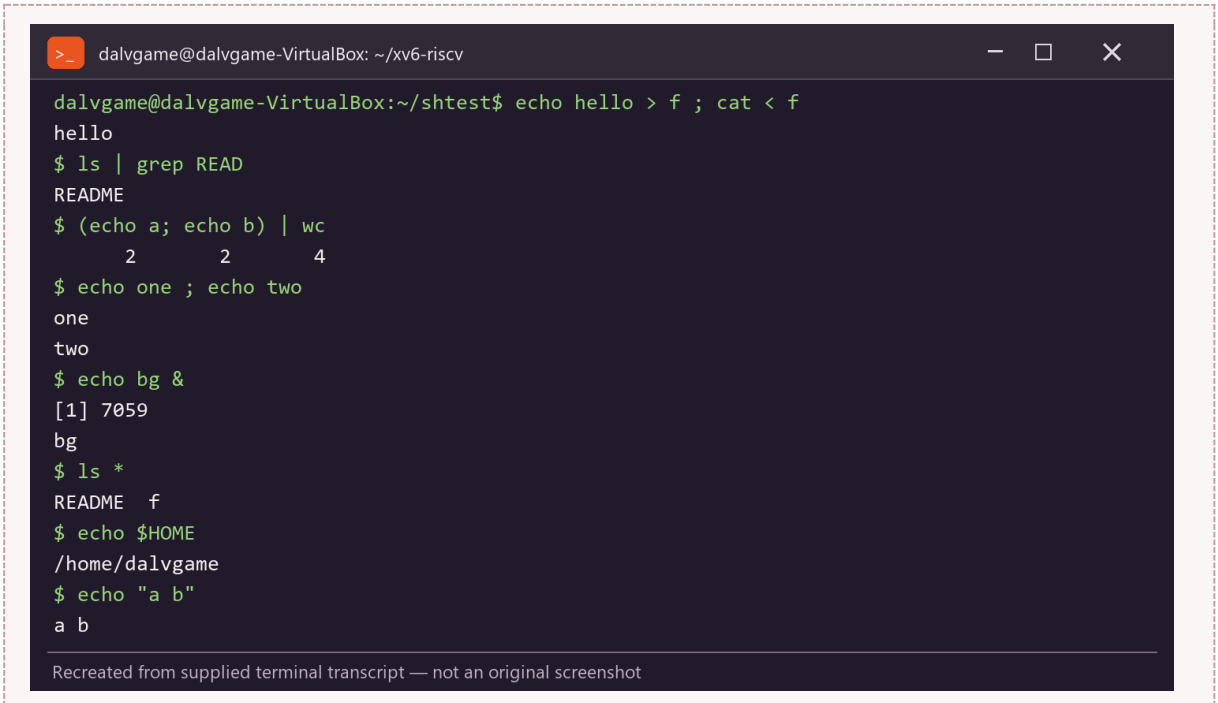


Figure 2.1 – The commands run in bash

Comment and observation

Everything worked in bash. `hello` was written to `f` and read back; the pipe passed the listing through `grep`; the group `(echo a; echo b)` produced two lines which `wc` counted as `2 2 4`; `;` ran the two `echo` commands one after the other; `&` started `echo bg` in the background (job `[1]`, later reported as `Done`); `ls *` listed `README` and `f`; `$HOME` became `/home/dalvgame`; the quotes of `"a b"` were removed; and after `cd d` the command `ls ..` still found the program and listed the parent folder.

2.2 The Same Commands in xv6

```
$ echo hello > f ; cat < f
hello
$ ls | grep READ
README          2 2 2441
$ (echo a; echo b) | wc
2 2 4
$ echo one ; echo two
one
two
$ echo bg &
bg
$ ls *
ls: cannot open *
$ echo $HOME
$HOME
$ echo "a b"
"a b"
$ mkdir d
$ cd d
$ ls ..
exec ls failed
```

The evidence for this exercise is shown in Figure 2.2.

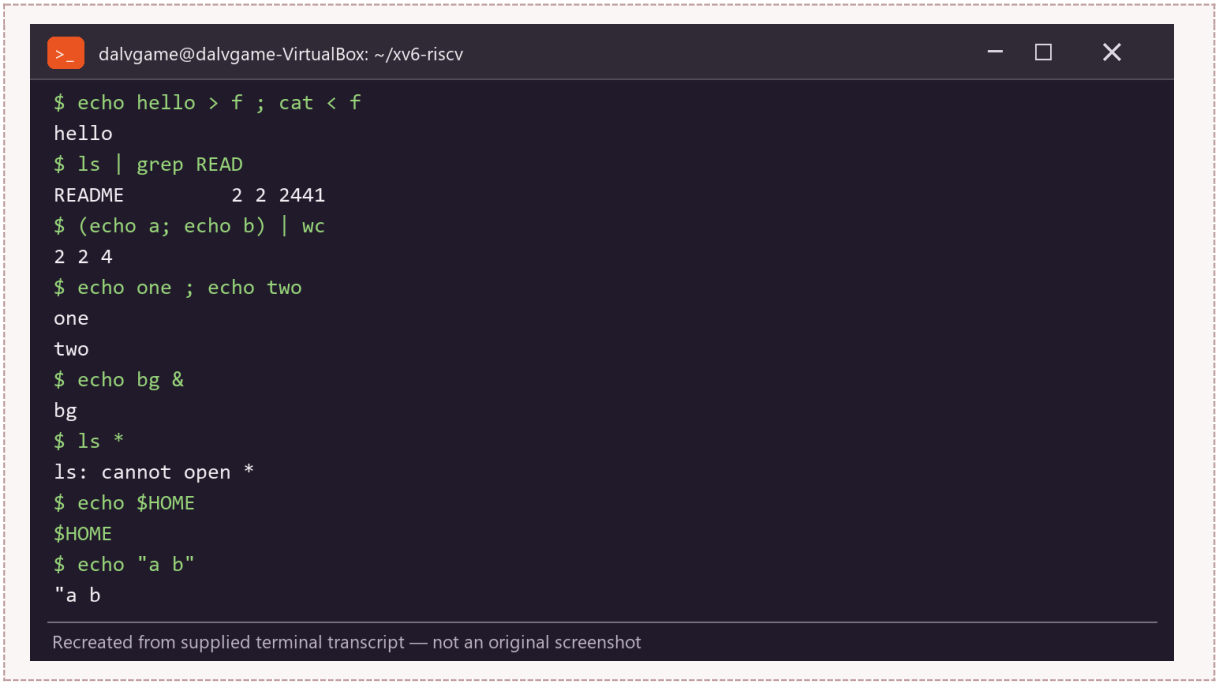


Figure 2.2 – The same commands run in xv6

Comment and observation

Redirection, the pipe, the group, the sequence and the background command work in the xv6 shell just as in bash. Globbing and variables do not: `ls *` looks for a file whose name is the single character `*`, and `echo $HOME` prints the text `$HOME`. Quotes are not removed, so `echo "a b"` prints the quote characters. The `ls` of xv6 prints one line per file (name, type, inode, size) and not a bare name, which is why `ls | grep READ` shows `README 2 2 2441`. After `cd d` the command `ls ..` fails (see 2.3).

One more difference was found by accident, when a Linux command line was typed at the xv6 prompt: `mkdir -p ~/shtest && cd ~/shtest && touch README` was answered with `leftovers: cd ~/shtest && touch README` and `syntax`. The xv6 shell has no `&&` (and no `||`) and no ~; the parser stopped at the unknown operator and reported a syntax error.

2.3 ls Failing after cd d, and the Alternative

```
$ mkdir d
$ cd d
$ ls ..
exec ls failed
$ /ls ..
(the listing of the root directory)
$ /ls .
.              1 27 32
..             1 1 1024
```

The evidence for this exercise is shown in Figure 2.3.



Figure 2.3 – ls failing after cd d, and the working alternative

Comment and observation

After `cd d` the shell is inside the folder `d`, and `ls ..` fails with `exec ls failed`. The xv6 shell has no PATH: it passes the bare name `ls` to `exec`, which looks for a file of that name in the current directory, and all xv6 programs are stored in the root directory `/`. Giving the full path, `/ls ..`, finds the program from any directory and prints the listing of the root (it also

shows the new folder `d` and the file `f` created earlier). `/ls .` shows the contents of the current folder `d`, which holds only `.` and `..`

2.4 The Comparison Table

Table 2.1 – bash and the xv6 shell, command by command

Command	Tests	bash	xv6
`echo hello > f ; cat < f`	redirection	hello	hello – the shell itself opens the files for < and >
`ls | grep READ`	pipe	README	`README 2 2 2441` – the shell joins two processes with a pipe; the xv6 `ls` prints name, type, inode, size
`(echo a; echo b) | wc`	grouping, list	2 2 4	2 2 4 – parentheses and `;` are parsed by the xv6 shell
`echo one ; echo two`	sequence	one, two	one, two
`echo bg &`	background	`[1] 7059`, bg, later `Done`	bg – the command runs in the background, but there is no job number and no Done message (no job control)
`ls *`	globbing	README f	`ls: cannot open *` – nobody expands `*`
`echo $HOME`	variables	/home/dalvgame	`$HOME` – the shell has no variables
`echo "a b"`	quotes	a b	`"a b"` – quotes are passed to the program as ordinary characters
`mkdir d ; cd d ; ls ..`	PATH	README d f	`exec ls failed` – no PATH, programs are looked up in the current directory only
arrow up, Tab, Ctrl+R	history, completion	work (line editing library)	do not work – the keys are echoed as text such as ^[[A; confirm with your own test

2.5 Answers to the Observe Questions

(1) Features of both shells and features of bash only. Both shells can run a program with arguments, redirect input and output (<, >), join programs with pipes (`|`), run lists of commands (`;`), group commands with parentheses, start commands in the background (`&`) and change the directory with the built-in `cd`. Only bash has globbing (`*`, `?`, `[ ]`), variables and their expansion, quote removal, a PATH search, job control (`jobs`, the `Done` message), `&&` and `||`, command history, Tab completion with line editing, and scripting constructs such as `if`, `for` and functions.

(2) Why does ls stop working after cd d? The xv6 shell has no PATH. It calls `exec("ls", ...)`, and `exec` opens a file with that name relative to the current directory. The programs of xv6 live in the root directory, so as soon as the current directory is `d`, there is no file `ls` there and `exec` fails. The way around is to give the path of the program: `/ls ..` (or `../ls ..`) runs it from inside `d`, as shown in Figure 2.3.

(3) Which missing feature would I miss most, and which is easiest to add? In daily work I would miss history with Tab completion most (together with globbing): every long command and every file name would have to be typed in full and any mistake retyped. The easiest feature to add is a PATH search: when `exec` fails, the shell would simply try the same name in a fixed

list of directories such as `/`, which is only a few lines in `runcmd`. History and completion are much harder, because the shell would have to read single keystrokes and handle the terminal escape sequences instead of reading whole lines.

3 WHO EXPANDS *? A PROGRAM THAT PRINTS ITS ARGV

This chapter corresponds to Part 3 of the handout. The program `args` prints the number of arguments it received and every argument inside square brackets, so nothing that happens to the command line before the program starts can stay hidden.

3.1 The Program args.c for xv6

```
#include "kernel/types.h"
#include "user/user.h"

int
main(int argc, char *argv[])
{
  printf("argc = %d\n", argc);
  for(int i = 0; i < argc; i++)
    printf("argv[%d] = [%s]\n", i, argv[i]);
  exit(0);
}
```

The line `$U/_args\` was added to `UPROGS` in the Makefile, right after `$U/_sleep\` (lines 143–145 of the Makefile hold `_sh`, `_sleep` and `_args`):

```
143:  $U/_sh\
144:  $U/_sleep\
145:  $U/_args\
146:  $U/_stressfs\
```

The evidence for this exercise is shown in Figure 3.1.



Figure 3.1 – args.c and the UPROGS line

Comment and observation

The program uses only the xv6 headers `kernel/types.h` and `user/user.h`. `argc` and `argv` are prepared by the `exec` system call from the list of strings that the shell passes to it; the program itself never sees the original command line.

3.2 args in xv6

```
$ args one two three
argc = 4
argv[0] = [args]
argv[1] = [one]
argv[2] = [two]
argv[3] = [three]
$ args *.c "x y"
argc = 4
argv[0] = [args]
argv[1] = [*.c]
argv[2] = ["x]
argv[3] = [y"]
$ args $HOME
argc = 2
argv[0] = [args]
argv[1] = [$HOME]
$ args 'single quoted'
argc = 3
argv[0] = [args]
argv[1] = ['single]
argv[2] = [quoted']
```

The evidence for this exercise is shown in Figure 3.2.

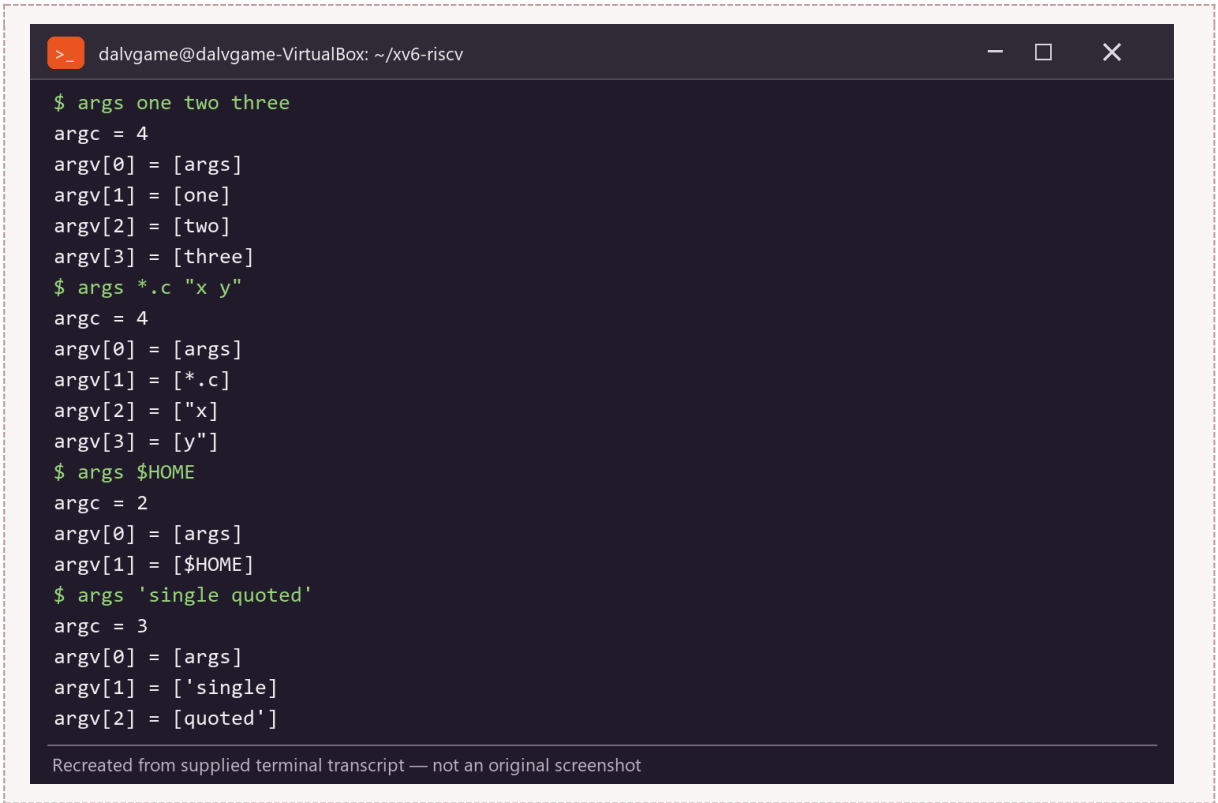


Figure 3.2 – args output in xv6

Comment and observation

In xv6 the words reach the program exactly as they were typed: `*.c` stays `*.c`, `$HOME` stays `$HOME`, and the quotes are kept as ordinary characters, so `"x y"` is cut into the two arguments `"x` and `y"`, and `'single quoted'` into `'single` and `quoted'`. A test with `args * "x y"` gave the same picture: the first argument was the single character `*`.

3.3 args_linux in bash

```
gcc -o args args_linux.c
./args one two three
./args *.c "x y"
./args $HOME
./args 'single quoted'
```

Output (the folder `~/lab4` contained `args.c`, `args_linux.c` and `sleep.c`):

```
argc = 4
argv[0] = [./args]
argv[1] = [one]
argv[2] = [two]
argv[3] = [three]
argc = 5
argv[0] = [./args]
argv[1] = [args.c]
argv[2] = [args_linux.c]
argv[3] = [sleep.c]
argv[4] = [x y]
argc = 2
argv[0] = [./args]
argv[1] = [/home/dalvgame]
argc = 2
argv[0] = [./args]
argv[1] = [single quoted]
```

The evidence for this exercise is shown in Figure 3.3.

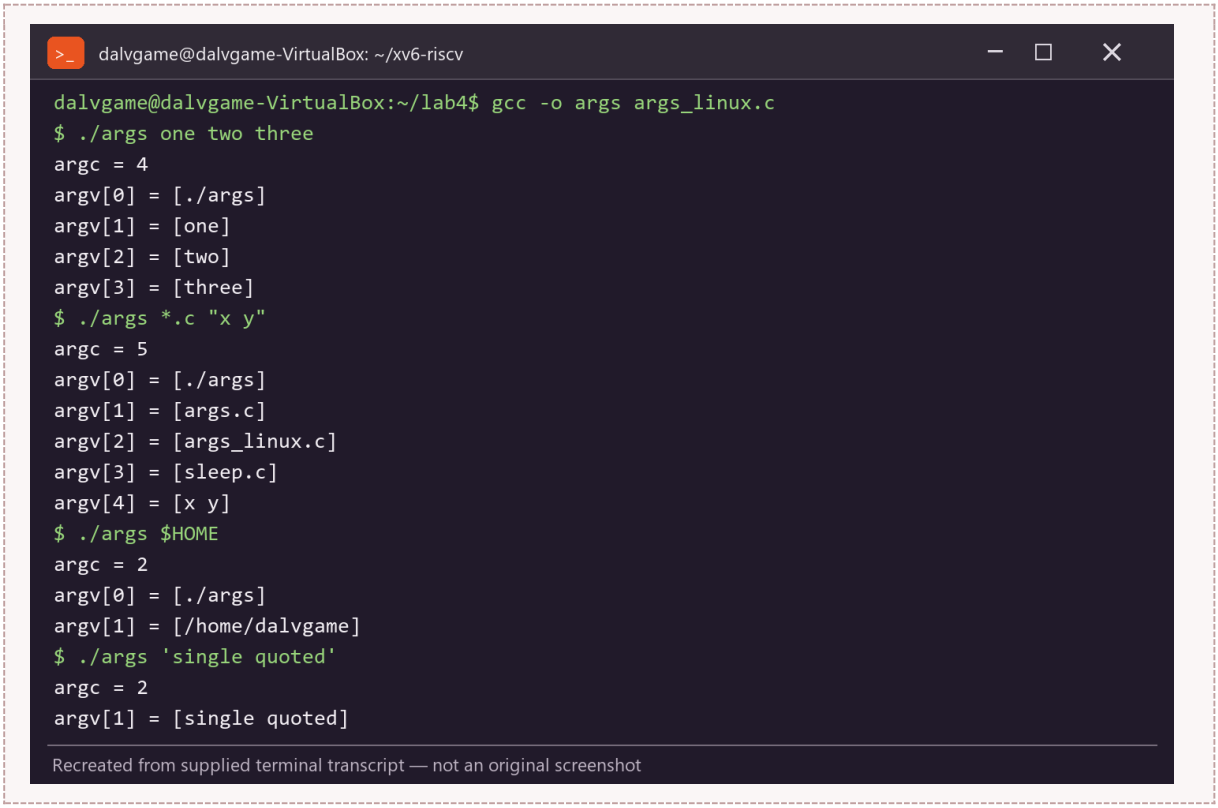


Figure 3.3 – args_linux output in bash

Comment and observation

The Linux version differs from the xv6 one only in the header (`<stdio.h>`) and in `return 0`. For `./args *.c "x y"` it received 5 arguments: the program name `./args`, the three file

names in alphabetical order, and the single argument `x y` with the quotes removed and the space kept. `$HOME` arrived as `/home/dalvgame`, and `'single quoted'` as one argument.

3.4 Answers to the Observe Questions

(1) Outputs for args *.c "x y". In bash (Figure 3.3) the program received 5 arguments**.** `./args`, `args.c`, `args_linux.c`, `sleep.c` and `x y`. In xv6 (Figure 3.2) it received 4: `args`, `*.c`, `"x` and `y"`. The difference is exactly the work that bash does before starting the program: expanding the pattern and removing the quotes.

(2) Who turns *.c into file names and removes the quotes? Neither the program nor the kernel: the shell does it. bash expands the pattern and strips the quotes first and only then call**`exec`**; the kernel merely copies the finished argument strings into the new process, and the program reads them from `argv`. The xv6 shell does not do either of these steps, which is why the program there gets the raw text.

(3) What happened to the extra spaces in one two three? They disappeared. The shell cuts the line into words at white space and passes only the words, so any number of spaces between two words is just a separator. Both `args one two three` and `args one two three` give the same `argv`, and this also holds in xv6 (the tokenizer skips white-space characters, see Section 4.4).

4 READING THE SHELL

This chapter corresponds to Part 4 of the handout. In `~/xv6-riscv` the commands of the handout were run and the parts of `user/sh.c` they point to were read.

4.1 The Kinds of Commands

```
wc -l user/sh.c
499 user/sh.c
grep -n "#define" user/sh.c
8:#define EXEC  1
9:#define REDIR 2
10:#define PIPE  3
11:#define LIST  4
12:#define BACK  5
14:#define MAXARGS 10
```

The evidence for this exercise is shown in Figure 4.1.



Figure 4.1 – The command kinds defined in sh.c

Comment and observation

The whole shell is 499 lines long. The `#define` lines name the five kinds of command the shell knows: `EXEC`, `REDIR`, `PIPE`, `LIST` and `BACK`. A sixth constant, `MAXARGS`, limits a simple command to 10 arguments.

4.2 Where Each Kind Is Run

```
grep -n "case " user/sh.c
75:  case EXEC:
83:  case REDIR:
93:  case LIST:
101:  case PIPE:
125:  case BACK:
284:  case 0:
286:  case '|':  ...  (lines 287-294: ( ) ; & < >)
391:    case '<':
394:    case '>':
397:    case '+': // >>
469:  case EXEC:  ...  (lines 475-493: REDIR PIPE LIST BACK)
```

The evidence for this exercise is shown in Figure 4.2.

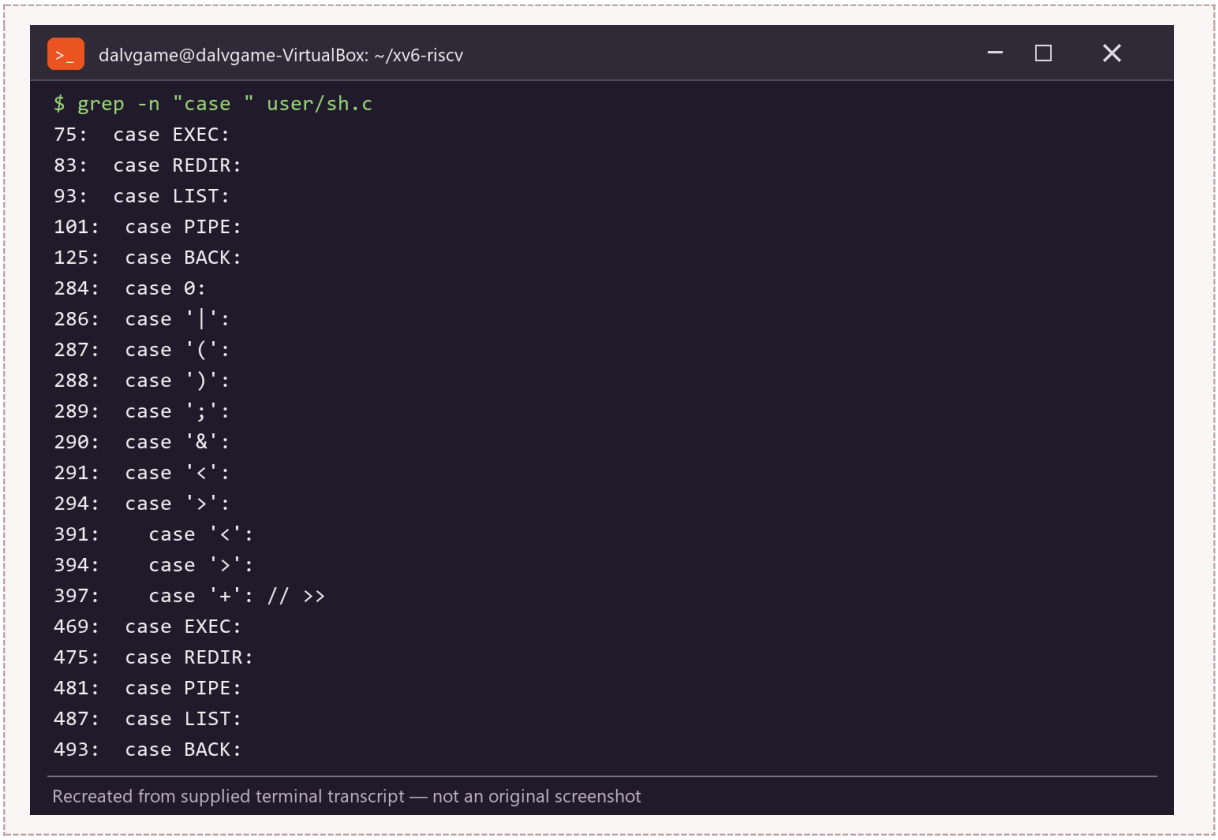


Figure 4.2 – Where each kind is run

Comment and observation

The `case` labels fall into three groups. In `runcmd` (lines 75 to 125) there is one branch per kind: run a program, open the redirection file, run both sides of a list, create a pipe and two children, run a command in the background. In `gettoken` (lines 284 to 294) the characters `|`, `(`, `)`, `;`, `&`, < and > are recognised as symbols, and in `parseredirs` (lines 391 to 397) the cases <, > and + handle the redirection operators, + being the second character of >>. Finally `nulterminate` (lines 469 to 493) walks the finished tree once more, again with one branch per kind.

4.3 The Code That Handles cd

```
grep -n "cd" user/sh.c | head
170:        fprintf(2, "cannot cd %s\n", cmd + 3);
```

The evidence for this exercise is shown in Figure 4.3.



Figure 4.3 – The code that handles cd

Comment and observation

Only one line of `sh.c` contains the text `cd`: the error message `cannot cd %s` on line 170, which belongs to the loop in `main` that reads a line with `getcmd`. Just above it (lines 161 to 164) the loop skips leading blanks and tests for an empty command; a line that starts with `cd` is then handled right there, with a call to `chdir` on the text after the third character (`cmd + 3`), and the loop goes on to the next line without any `fork`. Only the other lines are executed in a child process.

4.4 Answers to the Observe Questions

(1) The five kinds of command. `EXEC` – a program with its arguments (a plain word); `REDIR` – input or output redirection, produced by < and > (and >>); `PIPE` – two commands joined by `|`; `LIST` – commands run one after the other, produced by `;`; `BACK` – a command run in the background, produced by `&`. Parentheses only group a command and have no kind of their own.

(2) Where is cd handled and why is it not a separate program? In `main`, in the loop that reads a line, before the shell forks. The current directory is a property of a process. A separate program would run in a child process, change the directory of the child and exit, and the shell would still be in the old directory. Only the shell process itself can change its own directory, exactly as with the built-in `cd` of bash in Laboratory Work No. 3.

(3) Code that handles * or $. There is none. The search for `*` and `$` returned only lines in which `*` is the C pointer syntax (`char *argv[]`, `struct cmd *cmd`, casts such as `(struct execcmd *)cmd`), and the only `$` is the prompt string `"$ "` written by `getcmd` on line 137. The tokenizer knows only white space and the symbols `|`, `(`, `)`, `;`, `&`, <, >; everything else is a word that is passed on unchanged. This explains Part 2: there is no globbing and no variables, so `ls *` looks for a file called `*` and `echo $HOME` prints `$HOME`. The same tokenizer treats a quote as an ordinary character, which is why `args` showed `"x` and `y"` as two arguments, and it skips white space between words (lines 278 and 303), which explains why extra spaces disappear.

CONCLUSION

The `sleep` program was adapted to the current xv6 by replacing the call `sleep()` with `pause()`, the name declared on line 25 of `user/user.h`; it builds with `-Wall -Werror` and works with and without an argument.

The side-by-side comparison showed that a shell of 499 lines already provides the core of a Unix shell: programs with arguments, redirection, pipes, lists, grouping, background execution and a built-in `cd`. Everything that makes bash comfortable (globbing, variables, quote removal, `&&` and `||`, a PATH search, job control, history and completion) is added on top. The missing PATH explained why `ls` fails after `cd d` and why `/ls ..` works.

The program `args` made the division of work visible: in bash the shell turns `*.c` into file names and removes the quotes before `exec`, and the program receives the result (5 arguments), while in xv6 the program receives the text as typed (4 arguments). Reading `user/sh.c` confirmed it: the five kinds of command are the whole language, `cd` is handled by the shell itself because the working directory belongs to the shell process, and no code handles `*` or `$`.

REFERENCES

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[2] MIT PDOS. xv6-riscv source repository. https://github.com/mit-pdos/xv6-riscv.

[3] R. Cox, F. Kaashoek, R. Morris. xv6: a simple, Unix-like teaching operating system. MIT 6.1810. https://pdos.csail.mit.edu/6.1810/.

[4] Free Software Foundation. GNU Bash Reference Manual. https://www.gnu.org/software/bash/manual/.